

ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Department of IT, CSE(DS)

Synopsis On

SCHEME-AI

Government Scheme Eligibility & Application AI Assistant

1. INTRODUCTION

1.1 Overview

India operates thousands of welfare schemes introduced by Central and State Governments for different categories of citizens such as farmers, students, women, laborers, entrepreneurs, senior citizens, and economically weaker sections. These schemes provide financial assistance, healthcare support, insurance benefits, housing facilities, educational scholarships, pensions, employment opportunities, and social welfare services. Despite the availability of these schemes, a large number of eligible citizens are unable to receive benefits due to lack of awareness, fragmented information systems, language barriers, and complicated eligibility processes.

Most government portals are difficult for common citizens to understand because the information is spread across multiple websites and presented in complex administrative language. Citizens often struggle to identify whether they qualify for a scheme, what documents are required, where they should apply, and how to complete the application process successfully. Rural populations and digitally less-aware users face even greater difficulties because of limited technical literacy and lack of multilingual accessibility.

To address these challenges, this project proposes **Scheme-AI: Government Scheme Eligibility & Application AI Assistant**, an AI-driven intelligent platform that assists citizens in discovering and understanding government welfare schemes through conversational interaction. The system uses modern technologies such as Agentic AI, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), and Large Language Models (LLMs) to provide personalized scheme recommendations based on the user's profile including age, income, occupation, state, gender, education level, and social category.

The assistant simplifies complicated government information into plain language and guides users step-by-step through eligibility verification, required documents, registration procedures, deadlines, and official application portals. The platform also supports multilingual interaction, making it accessible for citizens across different regions of India.



Figure 1: Step-by-Step SCHEME-AI Process in India

1.2 Purpose

The primary purpose of the proposed system is to simplify access to government welfare schemes by providing an AI-powered platform that helps citizens identify, understand, and apply for schemes for which they are eligible. The system aims to eliminate information barriers, reduce manual effort, and improve awareness of welfare opportunities across different sections of society. The system focuses on the following objectives:

1. Centralize Government Scheme Information

The system collects and organizes information from multiple Central and State Government portals into a unified knowledge base. This enables users to access scheme details, eligibility criteria, benefits, required documents, and application procedures from a single platform

2. Automate Eligibility Identification

Using AI-based profiling and intelligent matching algorithms, the system analyzes user information such as age, gender, income, occupation, education level, state, and social category to identify schemes for which the user is eligible

3. Provide Personalized Scheme Recommendations

The assistant delivers customized recommendations based on individual user profiles. Instead of manually searching through numerous government websites, users receive a filtered list of relevant schemes suited to their needs and circumstances.

4. Simplify Application Guidance and Documentation

The system explains complex government procedures in simple language and provides step-by-step guidance regarding registration, required documents, application submission, and important deadlines, reducing confusion during the application process

5. Support Multilingual and Conversational Interaction

Through Natural Language Processing (NLP) and multilingual support, citizens can interact with the assistant in their preferred language. This improves accessibility for rural populations and users with limited technical knowledge

6.Improve Awareness and Accessibility of Welfare Programs

By making scheme information easily understandable and readily available, the platform increases public awareness and helps ensure that eligible citizens can access government benefits more effectively.



Figure 2: SCHEME-AI Tracking & Analytics Dashboard

By achieving these objectives, the proposed Scheme-AI platform simplifies access to government welfare schemes through AI-driven recommendations and automated eligibility assessment. The system enhances citizen awareness, reduces the complexity of scheme discovery and application procedures, and improves accessibility through multilingual support. Ultimately, it enables eligible beneficiaries to access government services more efficiently and transparently.

2. LITERATURE SURVEY

2.1 Existing Problem

The government welfare ecosystem in India consists of thousands of schemes offered by Central and State Governments. Although these schemes are designed to support different sections of society, citizens often face significant challenges in identifying and accessing the benefits relevant to them. Current approaches to scheme discovery and application exhibit several critical limitations:

1. Manual Information Search and Verification

Citizens are required to visit multiple government websites and portals to search for welfare schemes. They must manually verify eligibility criteria, compare benefits, and understand application procedures. This process is time-consuming, confusing, and often results in eligible beneficiaries missing important opportunities due to lack of awareness or incomplete information.

2. Existing Government Portal Limitations

Most government portals provide information in a static and document-centric format. Users are expected to interpret lengthy guidelines, eligibility conditions, and official notifications on their own. These platforms generally lack personalized recommendations, conversational assistance, and intelligent filtering mechanisms that could help citizens quickly identify suitable schemes

3. *Accessibility and Language Barriers*

A large number of citizens, particularly those in rural areas, face difficulties due to limited digital literacy and language constraints. Many scheme portals primarily use English or complex administrative terminology, making it difficult for users to understand scheme details, required documents, and application procedures accurately

4. *Lack of Personalized Recommendation Systems*

Current systems do not effectively analyze user-specific information such as age, income, occupation, education level, gender, state, and social category. As a result, users must manually determine whether they qualify for a scheme, increasing the possibility of errors and missed opportunities

5. *Complex Application Procedures*

Government scheme applications often involve multiple forms, document requirements, verification steps, and deadlines. Many applicants struggle to understand the process, resulting in incomplete submissions, delays, or rejection of applications

6. *Limited Real-Time Assistance and Guidance*

Most existing platforms do not provide interactive support for answering user queries or guiding applicants through the registration process. Citizens frequently rely on third parties or local service centers, which may increase costs and reduce accessibility

<i>Approach</i>	<i>Method</i>	<i>Limitations</i>
Rule-Based Systems	Match user details with predefined eligibility rules from government databases .	Inflexible, difficult to update with new schemes
Information Retrieval Systems	Retrieve schemes using keyword matching and search algorithms	Low personalization, irrelevant results
Classification Models	Use Decision Trees, Random Forests, or Naive Bayes to predict scheme eligibility	Depends on quality and quantity of training data
NLP-Based Chatbots	Process user queries using Natural Language Processing techniques .	Limited contextual understanding, recommendation accuracy .
Deep Learning Models	Neural Networks trained on citizen and scheme datasets for personalized recommendations	High computational cost, requires large datasets

Table 1: Comparison of Existing Welfare Scheme Recommendation Systems

2.2 Proposed Solution

The proposed AI-Powered Government Welfare Scheme Recommendation System addresses existing limitations through an integrated, end-to-end architecture combining automated data acquisition, welfare scheme knowledge extraction, machine learning-based eligibility prediction, personalized recommendation generation, and conversational assistance. The solution differentiates itself through intelligent scheme matching, real-time data synchronization, and citizen-centric guidance mechanisms that simplify access to government welfare benefits.

Comprehensive Welfare Data Aggregation Framework

Unlike conventional government portals that rely on fragmented information sources, the proposed system implements automated data collection and processing mechanisms to gather welfare scheme information from multiple official platforms and normalize it into a unified internal format. The framework periodically updates scheme details, eligibility criteria, benefits, required documents, application procedures, and policy notifications to ensure recommendations are generated using the latest available information.

These heterogeneous inputs are stored in a structured format and exposed to the Recommendation Engine as a feature set{Age, Income, Gender, Education, Occupation, State, Category, Disability_Status, Family_Size, Benefit_Type, Scheme_Category, Eligibility_Score,...}

Rather than requiring citizens to manually search through multiple welfare portals, the system applies machine learning and recommendation algorithms to evaluate user profiles against scheme eligibility criteria. The engine then identifies and ranks suitable welfare schemes based on eligibility, relevance, and expected benefits.

- 1.**Data.gov.in Datasets:** Welfare scheme data, beneficiary statistics, and demographic indicators from government open-data repositories.
- 2.**MyScheme Portal:** Scheme details, eligibility criteria, benefits, and application procedures.
- 3.**PM Scheme Websites:** Information from PM-KISAN, PMAY, Ayushman Bharat, National Scholarship Portal, and other flagship initiatives.
- 4.**State Government Portals:** State-specific welfare schemes, subsidies, and social assistance programs.
- 5.**Government Notifications:** Policy updates, new scheme announcements, revised eligibility rules, and application deadlines.

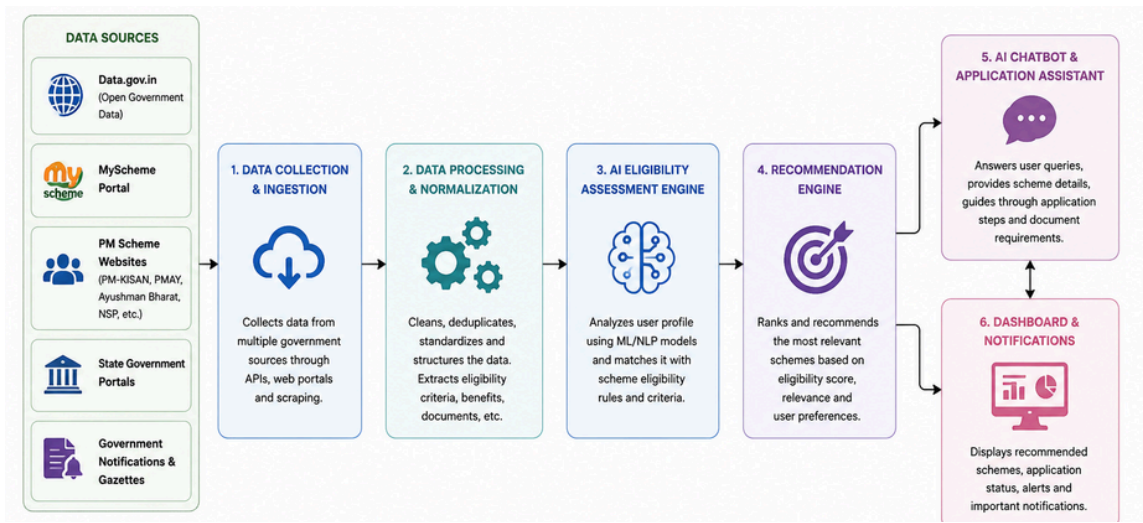


Figure 3: SCHEME-AI Decision System Architecture

The recommendation models are trained using welfare scheme datasets collected from Data.gov.in, MyScheme Portal, PM scheme websites, and state government portals. The dataset contains scheme eligibility parameters, beneficiary categories, demographic indicators, and application requirements. The data is divided into 70-20-10 train-validation-test sets. Cross-validation techniques help improve model performance and reduce overfitting, while hyperparameter tuning is performed to enhance recommendation accuracy.

Intelligent Recommendation Strategies

The system implements multiple recommendation modules that activate based on user profile attributes and eligibility outcomes:

<i>Strategy</i>	<i>Trigger Conditions</i>	<i>Recommendation Logic</i>
Eligibility Matching	User satisfies scheme eligibility criteria	Recommend schemes with highest eligibility score and benefit relevance
Benefit Prioritization	Multiple eligible schemes available	Rank schemes based on benefits, subsidies, and support value
Category-Based Recommendation	Multiple eligible schemes available	Suggest schemes from relevant categories
Assistance Recommendation	Multiple eligible schemes available	Recommend alternative schemes and application guidance

Table 2: Welfare Scheme Recommendation Strategy Matrix

The recommendation engine assigns scores to eligible schemes based on relevance, eligibility confidence, and potential benefits, ensuring that users receive the most suitable welfare programs according to their profile.

□ *Conversational AI Assistant and User Guidance*

The proposed system incorporates a conversational AI assistant powered by Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs). Citizens can interact with the platform using natural language queries to obtain scheme-related information without navigating complex government websites.

1. ***Eligibility Query Support:*** Users can ask whether they qualify for specific schemes based on their profile information.
2. ***Scheme Information Retrieval:*** The assistant provides details regarding benefits, eligibility criteria, required documents, and application procedures.
3. ***Application Guidance:*** Step-by-step instructions help users complete registrations and application submissions successfully.
4. ***Document Assistance:*** The system explains document requirements and verification processes for different welfare schemes.

The conversational interface reduces dependency on intermediaries and makes welfare information more accessible to citizens.

Multilingual Support and Accessibility Enhancement

The platform supports multilingual communication to improve accessibility across diverse regions of India.

1. ***Regional Language Support:*** Users can interact with the system in English, Hindi, and other regional languages.
2. ***Simplified Information Delivery:*** Government notifications and eligibility conditions are translated into easy-to-understand language.
3. ***Accessibility for Rural Users:*** The platform assists users with limited technical knowledge through guided interactions.
4. ***Inclusive User Experience:*** Citizens from different educational backgrounds can access welfare information without requiring specialized knowledge of government procedures

Real-Time Monitoring and Scheme Updates

The system continuously monitors government sources and updates its knowledge base to ensure recommendation accuracy.

1. ***Scheme Update Tracking:*** New scheme announcements, eligibility revisions, and policy changes are automatically incorporated.
2. ***Deadline Monitoring:*** Application deadlines and important dates are tracked and communicated to users.
3. ***Notification System:*** Users receive alerts regarding newly available schemes and relevant updates.
4. ***Recommendation Refresh:*** Eligibility matching is periodically recalculated to ensure recommendations remain current

□ ***System Deployment Architecture***

The platform supports flexible deployment options for development and production environments.

1. **Local Development Environment:** *The system can be executed on personal computers for testing, development, and experiment.*
2. **Cloud Deployment:** *Production deployment can be performed using cloud infrastructure to provide scalable and reliable access.*
3. **Database Integration:** Centralized storage maintains scheme information, user profiles, eligibility criteria, and recommendation history.
4. **API-Based Architecture:** External government data sources and services can be integrated through APIs for seamless information retrieval.

2.2.1 Conversational AI Assistant Integration

To enhance citizen interaction and simplify access to government welfare information, the proposed system integrates a Conversational AI Assistant powered by Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs). This integration enables users to communicate with the platform using natural language rather than manually searching through multiple government portals and documents.

The AI assistant acts as the primary interface between citizens and the welfare recommendation system. Users can ask questions regarding scheme eligibility, benefits, required documents, application procedures, and deadlines. The assistant processes these queries, retrieves relevant information from the centralized scheme database, and generates accurate, user-friendly responses.

The conversational layer operates in two major ways: information retrieval and personalized guidance. For information retrieval, the assistant accesses scheme repositories, government notifications, and eligibility databases to provide up-to-date responses. For personalized guidance, the system evaluates user profiles and recommends welfare schemes based on eligibility and benefit relevance.

A typical interaction flow may operate as follows:

1. The user provides personal information such as age, income, occupation, education level, state, and social category.
2. The Eligibility Engine analyzes the provided information and identifies matching welfare schemes.
3. The Recommendation Module ranks schemes according to eligibility confidence and potential benefits.
4. The AI Assistant presents personalized recommendations and explains eligibility requirements.
5. The user can request additional information regarding documents, application procedures, and deadlines.
6. The system provides step-by-step guidance to assist the user throughout the application process.

By integrating Conversational AI technologies, the proposed system improves accessibility, reduces information barriers, and enables citizens to access government welfare opportunities through an intuitive and user-friendly interface

2.2.2 Retrieval-Augmented Generation (RAG) Based Knowledge Retrieval System

The proposed system incorporates a Retrieval-Augmented Generation (RAG) framework to provide accurate, up-to-date, and context-aware responses regarding government welfare schemes. Instead of relying solely on pre-trained language models, the RAG architecture retrieves relevant information from a centralized welfare scheme knowledge base before generating responses.

The knowledge repository contains scheme descriptions, eligibility criteria, benefits, required documents, application procedures, government notifications, and policy updates collected from official sources. When a user submits a query, the retrieval module searches the knowledge base for the most relevant information and forwards the retrieved content to the language model.

The RAG framework offers several advantages over traditional chatbot systems. First, it reduces the possibility of inaccurate or outdated responses by grounding generated answers in verified government information. Second, it enables the assistant to incorporate newly added schemes and policy updates without requiring complete model retraining. Third, it improves transparency by ensuring recommendations are supported by retrieved scheme data.

A typical RAG workflow operates as follows:

1. The user submits a query related to government welfare schemes.
2. The Retrieval Engine searches the knowledge base for relevant scheme documents and policy information.
3. The most relevant content is selected and provided as contextual information.
4. The Language Model generates a personalized and easy-to-understand response based on the retrieved data.
5. The system presents scheme recommendations, eligibility details, required documents, and application guidance to the user.

By integrating Retrieval-Augmented Generation, the proposed platform delivers reliable, explainable, and up-to-date welfare assistance while maintaining the flexibility to accommodate future government scheme updates.

3. THEORETICAL ANALYSIS

3.1 Block Diagram

Workflow Explanation:

The system architecture comprises seven interconnected modules operating in a sequential yet interactive workflow to identify eligible government schemes, generate personalized recommendations, and assist citizens throughout the application process.

Workflow Explanation:

1. User Profile Input Module collects user information such as age, gender, income, occupation, education level, state, social category, disability status, and family details through an interactive interface.
2. Welfare Data Aggregation Module continuously gathers scheme-related information from government portals, public repositories, policy notifications, and welfare databases. The collected information includes eligibility criteria, benefits, required documents, application procedures, and deadlines.
3. Data Processing and Normalization Module validates collected information, removes inconsistencies, standardizes data formats, and converts scheme details into a structured representation suitable for analysis and retrieval.
4. Eligibility Assessment Module compares user profile attributes with scheme eligibility conditions. The module automatically determines whether a user qualifies for a particular welfare scheme and calculates an eligibility confidence score.
5. Recommendation Engine Module ranks eligible schemes based on relevance, benefits, eligibility confidence, and user requirements. The system generates personalized recommendations to help citizens identify the most suitable welfare opportunities.
6. Conversational AI and RAG Module retrieves relevant information from the knowledge base and generates user-friendly responses using Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs). The assistant answers user queries regarding scheme details, benefits, required documents, and application procedures.
7. Dashboard and Notification Module presents recommendations, eligibility results, scheme updates, and application guidance through an intuitive dashboard. The system also sends notifications regarding new schemes, policy changes, important deadlines, and application status updates.

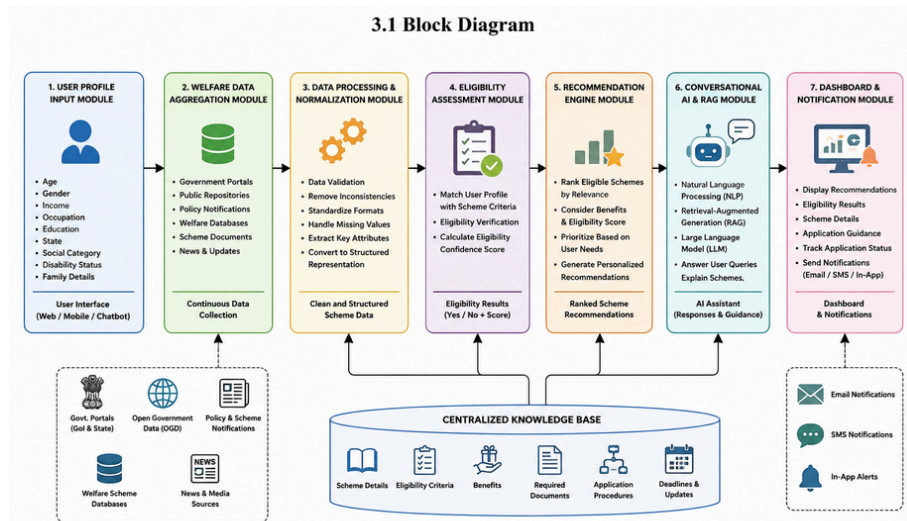


Figure 4: Block Diagram

The modules operate collaboratively to transform the traditionally complex process of government scheme discovery into an intelligent, automated, and citizen-friendly experience

3.2 Hardware/Software Designing

Hardware Requirements:

<i>Component</i>	<i>Minimum Specification</i>	<i>Recommended Specification</i>
Processor	Intel Core i3 (8th gen) or AMD Ryzen 5, 2.5 GHz, 4 cores	Intel Core i5 (10th gen+) or AMD Ryzen 7, 3.5 GHz, 8 cores
RAM	8 GB DDR4	16 GB DDR4 (for large dataset training)
Storage	256 GB SSD	512 GB NVMe SSD (faster I/O for database operations)
Network	Stable broadband, 10 Mbps	Fiber connection, 50+ Mbps (low latency for API calls)
Display	1366 x 768 resolution	Full HD 1920×1080

Table 3: Development and Testing Hardware

Software Requirements:

<i>Category</i>	<i>Software/Library (Version)</i>
Operating System	Ubuntu 22.04 LTS, Windows 10/11, macOS Monterey+
Programming Language	Python 3.10+ (3.11 recommended for performance improvements)
Data Manipulation	Pandas 2.0+, NumPy 1.24+, OpenPyXL (Excel handling)
Machine Learning	Scikit-learn 1.3+, XGBoost 2.0+, LightGBM 4.0+, TensorFlow 2.13+ (for LSTM), Optuna 3.3+ (hyperparameter tuning)
Data Collection	My Scheme , Data.gov.in , State portals , Ministry websites , Government notif
Natural Language Processing	Lang Chain , NLTK 3.8+, spaCy 3.6+, Transformers 4.31+ (BERT for sentiment)
Frontend	Streamlit , React
Database	My SQL , Mongo DB , SQLite (development), PostgreSQL 15+ (production)
Visualization	Matplotlib 3.7+, Plotly 5.16+, Dash 2.13+ (interactive dashboards), Streamlit 1.26+ (alternative UI)
Vector Database	ChramDB , FAISS
Deployment	Docker 24+, Docker Compose 2.20+, Nginx (reverse proxy), Supervisor (process management)
Version Control	Git 2.41+, GitHub/GitLabfor repository hosting
AI Model	Open GPT , Liama

Table 4: Software Stack and Dependencies

Development Environment Setup:

1. ***Virtual Environment:*** Use Python venv or conda to isolate dependencies and maintain a consistent development environment.
2. ***IDE:*** Visual Studio Code with Python extensions, PyCharm Professional (student license available), or Jupyter Notebook for exploratory data analysis.
3. ***Version Control:*** Use Git and GitHub for source code management, collaboration, and project tracking.
4. ***API Configuration:*** Configure API keys and external services used for LLM access, translation services, and government data retrieval.
5. ***Testing Framework:*** pytest for unit testing critical functions (data validation, order logic), backtest engine using historical data to simulate trades.

Security Considerations:

- ***User Data Protection:*** Store user information securely and avoid exposing sensitive personal data.
- ***Authentication & Authorization:*** Implement secure login and role-based access control where applicable.
- ***Encryption:*** Use HTTPS for all communications and encrypt sensitive data stored in databases.
- ***API Security:*** Protect API keys using environment variables and secret management practices.
- ***Data Integrity:*** Validate and sanitize all user inputs before processing.

Database Schema Design:

- ***User_Profile Table:*** user_id (PK), age, gender, income, occupation, education_level, state, social_category.
- ***Scheme_Details Table:*** scheme_id (PK), scheme_name, eligibility_criteria, benefits, required_documents, application_process, deadline.
- ***Recommendation Table:*** recommendation_id (PK), user_id (FK), scheme_id (FK), eligibility_score, recommendation_rank, timestamp.
- ***Query_History Table:*** query_id (PK), user_id (FK), user_query, assistant_response, timestamp

- ***Scheme_Updates Table:*** update_id (PK), scheme_id (FK), update_type, update_description, update_date .

This schema supports efficient storage and retrieval of user profiles, welfare scheme information, recommendation results, conversational interactions, and scheme updates while ensuring scalability and maintainability of the proposed SCHEME-AI platform

4. APPLICATIONS

The proposed SCHEME-AI platform can be utilized across various domains to improve citizen access to government welfare programs and simplify the scheme application process. Major applications include:

1. Welfare Scheme Discovery

The system helps citizens identify government schemes that match their eligibility profile without manually searching through multiple portals.

2. Automated Eligibility Verification

Users can quickly determine whether they qualify for specific schemes based on factors such as age, income, occupation, education, state, and social category.

3. Personalized Recommendation System

The platform provides customized scheme recommendations, ensuring that citizens receive information relevant to their needs and circumstances.

4. Application Guidance and Support

The AI assistant provides step-by-step guidance regarding required documents, registration procedures, application submission, and important deadlines.

5. Rural and Multilingual Accessibility

The system enables users from different regions to access scheme information through multilingual conversational interfaces, improving accessibility for rural populations.

6. Government Information Assistance

Citizens can obtain instant answers regarding scheme benefits, eligibility conditions, policy updates, and application procedures through natural language interaction.

7. Digital Governance and Citizen Services

The platform can support government initiatives aimed at increasing welfare awareness, improving service delivery, and enhancing citizen engagement.

8. *Educational and Awareness Programs*

Educational institutions, NGOs, and social organizations can use the system to spread awareness about welfare opportunities available to different beneficiary groups.

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